

voltage_divider_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-05-18 | jlcpb

Board Summary

Property	Value
Layers	2 copper (F.Cu, B.Cu)
Footprints	4 (0 SMD, 0 THT, 4 other)
Nets	3
Traces	29 segments
Vias	6
Board Size	30.0 x 25.0 mm

Design Overview

Theory of Operation

Voltage Divider Test

Simple 2-resistor voltage divider

5V -> 2.5V (10k/10k)

Power Architecture

Power Rails: +5V, GND, PWR_FLAG

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
IN	PinHeader_1x02_P2.54mm_Vertical	1	J1
OUT	PinHeader_1x02_P2.54mm_Vertical	1	J2
10k	R_0805_2012Metric	2	R1, R2

DRC Status

Metric	Count
Errors	1
Warnings	0
Blocking	1

Status: FAIL ### Violations by Type

Violation Type	Count
via_in_pad	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 1 blocking DRC violations (via_in_pad (1))
- **[OPTIONAL]** Verify zone fill in KiCad for 1 zone-connected nets
- **[OPTIONAL]** Add zone for GND on appropriate copper layer

Routing Status

Metric	Value
Signal Net Completion	66.7% (2/3)
Overall Completion	66.7%
Complete Nets	2 / 3
Incomplete Nets	1
Unconnected Pads	2

Unrouted Signal Nets

- GND

Unrouted Signal Nets

- GND

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~0.55 USD
Components (estimated)	~0.21 USD
Assembly (estimated)	~1.93 USD
Total (estimated)	~2.69 USD
Batch Quantity	5
Batch Total (estimated)	~13.44 USD